

Week 5 Project Summary

HealthConnect Project Kickoff and Problem Understanding

Experience Lab Internship Programme

Prepared by: Ayden Demanou, Junior Data Scientist, AnalystLab Africa Consulting

1. What I Planned to Accomplish in Week 5

As per the Week 4 proposed focus, my Week 5 goal was to move from problem definition into practical modelling: prepare the dataset for machine learning, engineer features informed by the Week 4 EDA findings, implement a leakage-safe train/test split, build a baseline logistic regression model, and evaluate it with standard classification metrics.

2. What I Actually Completed

- Prepared the modelling dataset: filtered to Attended/No-Show records (excluding Cancelled per the Week 4 target definition), handled missing values, dropped irrelevant/leaky fields, and encoded categorical variables.
- Produced six decision-supporting visualisations (target distribution, distribution/outlier checks, correlation heatmap, feature-target relationship plot) and used each to justify a specific preprocessing or modelling choice.
- Engineered four new features: *previous_no_show_rate*, *is_new_patient*, *has_reminder*, and a consolidated *gender_grp* category.
- Implemented and verified a patient-grouped train/test split (zero patient overlap between sets).
- Built, trained, and evaluated a baseline logistic regression model.
- Documented cross-track collaboration with the Data Analytics track and updated the Week 4 assumptions/limitations/risks register.

3. Key Findings and Development Outcomes

- Baseline model: 62.7% accuracy and ROC-AUC of 0.68, meaningfully above the 50% naive majority-class baseline, with reasonably balanced precision (0.62) and recall (0.65) across both classes.
- Feature importance confirmed the Week 4 EDA: *booking_lead_days* and *previous_no_shows* are the two strongest drivers of predicted no-show risk.
- Discovered multicollinearity between *previous_appointments*, *previous_no_shows*, and the newly engineered *previous_no_show_rate* (correlation of 0.46 between *previous_appointments* and *previous_no_shows*), which produced a counter-intuitive negative coefficient on *previous_no_show_rate* — flagged for simplification in Week 6 rather than over-interpreted.
- Confirmed *waiting_time_minutes* has effectively no relationship with the outcome, supporting its exclusion as a leakage risk.

4. Major Challenges Encountered

- Deciding how to represent “prior patient behaviour” without introducing multicollinearity , previous_appointments, previous_no_shows, and previous_no_show_rate all overlap in what they measure, and untangling their individual contribution to the model was not straightforward with a linear baseline.
- Balancing thoroughness with the Week 5 scope: the brief provides several optional visualisation types (histograms, box plots, heatmaps, feature-target plots); I selected only the ones that changed a decision, per the instruction to avoid visualisations included purely to meet a count.

5. Important Decisions Made and Why

- Excluded Cancelled appointments from the primary binary target, keeping the Week 4 decision unchanged, since they remain a small (5.3%), behaviourally distinct group.
- Chose a patient-grouped split over a plain random split, to prevent the same patient's appointments leaking between training and test data
- Did not apply class-weighting or resampling, since the target remains close to balanced on the modelling subset (51.4%/48.6%).
- Kept previous_no_show_rate in the model despite its weak/negative coefficient, rather than removing it, since the issue is multicollinearity (documented) rather than the feature being meaningless

6. Changes to My Week 4 Approach

- Dropped age_group as a feature (redundant with age) — a minor refinement, not a reversal of the Week 4 plan.
- Grouped the rare gender category “Prefer not to say” (~2% of records) into “Other” before encoding, to avoid an unstable dummy variable — also a refinement rather than a change in modelling direction.
- No other Week 4 decisions (target definition, feature list, modelling approach, split strategy) were reversed; all were implemented as planned.

7. Cross-Track Collaboration Completed

Documented an interaction with the **Data Analytics track**: the no-show-rate-by-lead-time pattern found through EDA in this notebook mirrors the kind of KPI the Data Analytics track would independently calculate (no-show rate by scheduling dimension). Two tracks reaching the same conclusion by different methods increases confidence in the finding and was used to justify prioritising booking_lead_days as the leading model feature. This is also flagged as a ready hand-off point for the Machine Learning Engineering track, who can take the finalised feature list and preprocessing steps directly into their Week 5/6 pipeline work.

8. Remaining Work

- Resolving the multicollinearity between the three patient-history features.
- A fairness/bias check across gender and age subgroups on predicted risk, before considering any operational use of the model.

9. Proposed Focus for Week 6

- Train and compare tree-based ensemble models (Random Forest, Gradient Boosting) against the logistic regression baseline.
- Replace the single train/test split with grouped cross-validation (GroupKFold) for a more robust performance estimate.
- Resolve the multicollinearity among patient-history features.
- Run a fairness/bias check across demographic subgroups.
- Convert model output into an operational risk tier (Low/Medium/High) for use by the Data Analytics and Project Management tracks.